

Plan

- Vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$
- The Euclidean Space $\mathbb{R}^n$
- Vector Space
- Subspace
- Linear Combination
- Linear Dependence and Linear Independence
- Basis and Dimension

Vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$

Recall:

- A **vector** is a directed line segment - corresponds to a displacement from one point A to another B .

- A **vector** in $\mathbb{R}^2$:

$$\mathbf{v} = a\hat{\mathbf{i}} + b\hat{\mathbf{j}} \equiv \text{the position vector of the point } (a, b) := [a, b].$$

- A **vector** in $\mathbb{R}^3$:

$$\begin{aligned}\mathbf{v} &= a\hat{\mathbf{i}} + b\hat{\mathbf{j}} + c\hat{\mathbf{k}} \equiv \text{the position vector of the point } (a, b, c) \\ &:= [a, b, c].\end{aligned}$$

- Thus, the vector $[a, b, c]$ is identified by the position vector of (a, b, c) in $\mathbb{R}^3$.

Sometimes, we write the vector also as $\begin{bmatrix} a \\ b \\ c \end{bmatrix} = [a, b, c]^T$.

New Vectors from Old

- How do you add two vectors $\mathbf{u}$ and $\mathbf{v}$?

Using the ‘head-to-tail rule’ (which is same as the ‘parallelogram rule’), right?

- Suppose $\mathbf{u} = a_1 \hat{\mathbf{i}} + b_1 \hat{\mathbf{j}} + c_1 \hat{\mathbf{k}}$ and $\mathbf{v} = a_2 \hat{\mathbf{i}} + b_2 \hat{\mathbf{j}} + c_2 \hat{\mathbf{k}}$.
Then it follows from the ‘parallelogram rule’ that

$$\mathbf{u} + \mathbf{v} = (a_1 + a_2) \hat{\mathbf{i}} + (b_1 + b_2) \hat{\mathbf{j}} + (c_1 + c_2) \hat{\mathbf{k}}.$$

- Suppose k is a constant. What is the vector which is k times $\mathbf{u} = a_1 \hat{\mathbf{i}} + b_1 \hat{\mathbf{j}} + c_1 \hat{\mathbf{k}}$? Naturally,

$$k\mathbf{u} = ka_1 \hat{\mathbf{i}} + kb_1 \hat{\mathbf{j}} + kc_1 \hat{\mathbf{k}}.$$

The Euclidean Space $\mathbb{R}^n$

Definition: Let $n \in \mathbb{N}$. The space $\mathbb{R}^n$, as defined below, is called the n -dimensional **Euclidean space**.

$$\mathbb{R}^n = \{[x_1, x_2, \dots, x_n]^t : x_1, x_2, \dots, x_n \in \mathbb{R}\}.$$

- Elements of $\mathbb{R}^n$ are called **n -vectors** or simply **vectors**.
- Thus $\mathbb{R}^n$ is n -times Cartesian Product of $\mathbb{R}$.

- Note that $[x_1, x_2, \dots, x_n]^t = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ is a column vector.

- Sometimes, an element $[x_1, x_2, \dots, x_n]^t$ of $\mathbb{R}^n$ is also written as a row vector $[x_1, x_2, \dots, x_n]$ or $(x_1, x_2, \dots, x_n)$.
- The element $(x_1, x_2, \dots, x_n)$ is also termed as an n -tuple.
- The vector $[0, 0, \dots, 0]^t$ of $\mathbb{R}^n$, called the zero vector, is denoted by the symbol $\mathbf{0}$.
- If A is an $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$, then a solution of $A\mathbf{x} = \mathbf{b}$, if any, is an element of $\mathbb{R}^n$.
- Normally, while discussing a system of linear equations, elements of $\mathbb{R}^n$ are regarded as column vectors.
- Elements of $\mathbb{R}^n$ may also be regarded as row vectors.

Some Basic Properties:

Let $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and $c, d \in \mathbb{R}$. Then

- 1 $\mathbf{u} + \mathbf{v} \in \mathbb{R}^n$;
- 2 $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (commutativity);
- 3 $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$ (associativity);
- 4 $\mathbf{u} + \mathbf{0} = \mathbf{u}$;
- 5 $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$, where $-\mathbf{u} = (-1)\mathbf{u} = [-u_1, -u_2, \dots, -u_n]^T$;
- 6 $c\mathbf{u} \in \mathbb{R}^n$;
- 7 $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$ (distributivity over vector addition);
- 8 $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$ (distributivity over scalar addition);
- 9 $c(d\mathbf{u}) = (cd)\mathbf{u}$;
- 10 $1\mathbf{u} = \mathbf{u}$;

These properties are sufficient to do vector algebra in $\mathbb{R}^n$.

- If $\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{0} \in \mathbb{C}^n$ and $c, d \in \mathbb{R}$, we get all the previous ten properties.
- If $\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{0} \in \mathbb{C}^n$ and $c, d \in \mathbb{C}$, we get all the previous ten properties.
- If $A, B, C, \mathbf{O} \in M_2(\mathbb{C})$ (set of all 2×2 real matrices) and $c, d \in \mathbb{R}$, we get all the previous ten properties.
- If $p(x), q(x), r(x), \mathbf{0} \in \mathbb{R}_2[x]$ (set of all polynomials of degree at most two with real coefficients) and $c, d \in \mathbb{R}$, we get all the previous ten properties.

- The symbol $\mathbb{F}$ will be used as a representative of $\mathbb{R}$ or $\mathbb{C}$. That is, $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$.

Definition: Let $V (\neq \emptyset)$ be a set. For every $\mathbf{u}, \mathbf{v} \in V$ and $c \in \mathbb{F}$, let

- $\mathbf{u} + \mathbf{v}$ (called the vector addition) be defined; and
- $c \cdot \mathbf{u}$ (called the scalar multiplication) be defined.

Then V is called a vector space over $\mathbb{F}$ if

for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and for all $c, d \in \mathbb{F}$,

the following properties are satisfied:

- 1 $\mathbf{u} + \mathbf{v} \in V$;
- 2 $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$;
- 3 $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$;
- 4 There is an element $\mathbf{0}$, called a zero, such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$;
- 5 For each $\mathbf{u} \in V$, there is an element $-\mathbf{u}$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$;
- 6 $c.\mathbf{u} \in V$;
- 7 $c.(\mathbf{u} + \mathbf{v}) = c.\mathbf{u} + c.\mathbf{v}$;
- 8 $(c + d).\mathbf{u} = c.\mathbf{u} + d.\mathbf{u}$;
- 9 $c.(d.\mathbf{u}) = (cd).\mathbf{u}$; and
- 10 $1.\mathbf{u} = \mathbf{u}$.

- The elements of V are called **vectors**.
- The elements of $\mathbb{F}$ are called **scalars**.

Example

For any $n \geq 1$, the set $\mathbb{R}^n$ is a vector space over $\mathbb{R}$ with respect to usual operations of addition and scalar multiplication.

Example

For any $n \geq 1$, the set $\mathbb{C}^n$ is a vector space over $\mathbb{R}$ with respect to usual operations of addition and scalar multiplication.

Example

For any $n \geq 1$, the set $\mathbb{C}^n$ is a vector space over $\mathbb{C}$ with respect to usual operations of addition and scalar multiplication.

Example

*The set $\mathbb{R}^n$ is **not** a vector space over $\mathbb{C}$ with respect to usual operations of addition and scalar multiplication.*

Example

The set $\mathbb{Z}$ is *not* a vector space over $\mathbb{R}$ with respect to usual operations of addition and scalar multiplication.

Example

Let $\mathbb{R}_2[x] = \{a + bx + cx^2 : a, b, c \in \mathbb{R}\}$. For $p(x) = a_0 + b_0x + c_0x^2$, $q(x) = a_1 + b_1x + c_1x^2 \in \mathbb{R}_2[x]$ and $k \in \mathbb{R}$, define

$$p(x) + q(x) = (a_0 + a_1) + (b_0 + b_1)x + (c_0 + c_1)x^2$$

$$k.p(x) = (ka_0) + (kb_0)x + (kc_0)x^2.$$

Then $\mathbb{R}_2[x]$ is a vector space over $\mathbb{R}$.

Example

The set $M_2(\mathbb{R})$ of all 2×2 real matrices is a vector space over $\mathbb{R}$ with respect to usual operations of matrix addition and matrix scalar multiplication.

Example

The set $\mathbb{R}^2$ is *not* a vector space over $\mathbb{R}$ with respect to usual operations of addition and the following definition of scalar multiplication:

$$c \cdot [x, y]^t = [cx, 0]^t \quad \text{for } [x, y]^t \in \mathbb{R}^2, c \in \mathbb{R}.$$

- If there is no confusion, $c \cdot u$ is simply written as cu .

- We write $V(\mathbb{F})$ to denote that V is a vector space over $\mathbb{F}$.
- If $\mathbb{F}$ needs to be specified, then we write $V(\mathbb{F})$ to denote that V is a vector space over $\mathbb{F}$.
- We call V a **real vector space** or **complex vector space** according as $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$.

Result

Let V be a vector space over $\mathbb{F}$. Let $\mathbf{u} \in V$ and $c \in \mathbb{F}$.

- 1 $0.\mathbf{u} = \mathbf{0};$
- 2 $c.\mathbf{0} = \mathbf{0};$
- 3 $(-1).\mathbf{u} = -\mathbf{u};$ and
- 4 *If $c.\mathbf{u} = \mathbf{0}$ then either $c = 0$ or $\mathbf{u} = \mathbf{0}$.*

Subspace: Let V be a vector space and $(\emptyset \neq) W \subseteq V$. Then W is called a **subspace** of V if and only if $au + bv \in W$ for every $u, v \in W$ and for every $a, b \in \mathbb{F}$.

- If W is a subspace of a vector space V then $0 \in W$.
- The sets $\{0\}$ and V are always subspaces of any vector space V . These are called the **trivial** subspaces.
- If W is a subspace, and $u, v \in W$ then $au = au + 0u \in W$ and $u + v = 1u + 1v \in W$.
- W is a subspace iff W is closed under addition and scalar multiplication.
- If W is a subspace of a vector space V , then W is also a vector space.

Example

Examine whether the sets

$S = \{[x, y, z]^t \in \mathbb{R}^3 : x = y + 1\}$, $T = \{[x, y, z]^t \in \mathbb{R}^3 : x = 5y\}$
and $U = \{[x, y, z]^t \in \mathbb{R}^3 : x = z^2\}$ are subspaces of $\mathbb{R}^3$.

Example

Let $W = \{[x, y, z]^t \in \mathbb{R}^3 : x + y - z = 0\}$. Then W is a subspace of $\mathbb{R}^3$.

Example

Let W be the set of all 2×2 real symmetric matrices. Then W is a subspace of $M_2(\mathbb{R})$.

Example

Let A be an $m \times n$ matrix. Then $U = \{\mathbf{x} \in \mathbb{C}^n : A\mathbf{x} = \mathbf{0}\}$ is a subspace of $\mathbb{C}^n$, called the *null space* of A .

Example

Let U and W be two subspaces of V . Then

$U + W = \{\mathbf{u} + \mathbf{w} : \mathbf{u} \in U, \mathbf{w} \in W\}$ is also a subspace of V .

[If U and W are subspaces of V such that $U \cap W = \{\mathbf{0}\}$, then $U + W$ is called a **direct sum**. Notation: $U \oplus W$.]

Result

Let V be a vector space.

- If U and W are subspaces of V , then $U \cap W$ is also a subspace of V .
- If $\{U_i\}$ is a non-empty collection of subspaces of V , then $\bigcap_i U_i$ is also a subspace of V .
- Let U and W be subspaces of V . Then $U \cup W$ is a subspace of V iff $U \subseteq W$ or $W \subseteq U$.

An Example Revisited:

Consider the homogeneous system $A\mathbf{x} = \mathbf{0}$, where

$$A = \begin{bmatrix} 1 & -1 & -1 & 2 \\ 2 & -2 & -1 & 3 \\ -1 & 1 & -1 & 0 \end{bmatrix}, \quad \textcolor{blue}{RREF}(A) = \begin{bmatrix} 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The solution set for $A\mathbf{x} = \mathbf{0}$ is

$$S_h = \left\{ s \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 1 \\ 1 \end{bmatrix} : s, t \in \mathbb{C} \right\}.$$

Can we describe S_h with a few of the solutions? Can we derive some special properties of solution sets like S_h ?

Linear Combination:

A vector $\mathbf{v}$ in V is a **linear combination** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ in V if there exist $c_1, c_2, \dots, c_k \in \mathbb{F}$ such that

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k.$$

- The numbers $c_1, c_2, \dots, c_k$ are called the **coefficients** of the linear combination.

Example

Is the vector $[1, 2, 3]^t$ a linear combination of $[1, 0, 3]^t$ and $[-1, 1, -3]^t$?

Result

*A system of linear equations with augmented matrix $[A \mid \mathbf{b}]$ is consistent **if and only if** $\mathbf{b}$ is a linear combination of the columns of A .*

Span of a Set: Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be a subset of a vector space V . Then the set of all linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ is called the **span** of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$, and is denoted by $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k)$ or $\text{span}(S)$. That is,

$$\text{span}(S) = \{c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k \mid c_1, c_2, \dots, c_k \in \mathbb{F}\}.$$

Let $S \subseteq V$ (may be infinite!) The **span** of S is defined by

$$\text{span}(S) := \left\{ \sum_{i=1}^m \alpha_i \mathbf{v}_i \mid \mathbf{v}_i \in S, \alpha_i \in \mathbb{F}, m \text{ a nonnegative integer} \right\}.$$

- If $\text{span}(S) = V$, then S is called a **spanning set** for V .
- Convention: $\text{span}(\emptyset) = \{\mathbf{0}\}$

- For example, $\mathbb{R}^2 = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$, where $\mathbf{e}_1 = [1, 0]^t$ and $\mathbf{e}_2 = [0, 1]^t$.
- $\mathbb{R}_2[x] = \text{span}(1, x, x^2)$.
- $\mathbb{R}[x] = \text{span}(1, x, x^2, \dots)$ [$\mathbb{R}[x]$ = set of all polynomials in x].

Example

Let $\mathbf{u} = [1, 2, 3]^t$ and $\mathbf{v} = [-1, 1, -3]^t$. Describe $\text{span}(\mathbf{u}, \mathbf{v})$ in $\mathbb{R}^3$ geometrically.

Result

Let S be a subset of a vector space V . Then $\text{span}(S)$ is a *subspace* of V .

Linear Dependence: A set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ of vectors in a vector space V is said to be **linearly dependent (LD)** if there are scalars $c_1, c_2, \dots, c_k$, **at least one of them non-zero**, such that $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k = \mathbf{0}$.

An **infinite set** $S \subseteq V$ is **linearly dependent (LD)** if there is some **finite** linearly dependent subset of S .

Linear Independence: The set S of vectors in a vector space V is said to be **linearly independent (LI)** if it is **not** linearly dependent. Thus if $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ then S is **linearly**

independent (LI) if

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k = \mathbf{0} \Rightarrow c_1 = c_2 = \dots = c_k = 0;$$

if S is **infinite** then S is **linearly independent (LI)** if every **finite subset** of S is linearly independent.

- Any superset of a LD set is LD.
- Any subset of a LI set is LI.

- A set $S \subseteq V$ is linearly dependent iff there is $v \in S$ such that $v \in \text{span}(S \setminus \{v\})$, i.e., $\text{span}(S \setminus \{v\}) = \text{span}(S)$.
- Set $\{\mathbf{0}\}$ is linearly dependent as $1 \cdot \mathbf{0} = \mathbf{0}$. [A non-trivial linear combination of $\mathbf{0}$ is $\mathbf{0}$.] OR because $\mathbf{0} \in \text{span}(\emptyset)$.
- If $\mathbf{0} \in S$, then S is always linearly dependent as S contains a LD set $\{\mathbf{0}\}$.
- If $\{\mathbf{u}, \mathbf{v}\}$ is linearly dependent, then one of them is a scalar multiple of the other.
- The set $\emptyset$ is linearly independent. Convention?

Example

Let $\mathbf{e}_i \in \mathbb{R}^n$ be the i -th column of the identity matrix I_n . Is $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ linearly independent?

Example

The set $\{1, x, x^2, \dots, x^n\}$ is linearly independent in $\mathbb{R}_n[x]$.

Example

The set $\{1, x, x^2, \dots\}$ is linearly independent in $\mathbb{R}[x]$.

Example

The set $\{A, B, C\}$ is linearly dependent in $M_2(\mathbb{R})$, where

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}, C = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \text{ and } D = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Result

Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ be non-zero vectors in a vector space V .

Then they are linearly dependent *iff there is an integer r such that $\mathbf{v}_r$ is a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_{r-1}$, and $\mathbf{v}_1, \dots, \mathbf{v}_{r-1}$ are linearly independent.*

Example

Examine whether the following sets are linearly dependent.

- $S = \{[1, 4]^t, [-1, 2]^t\}$
- $T = \{[1, 2, 0]^t, [1, 1, -1]^t, [1, 4, 2]^t\}$

Linear Combinations of Rows

Suppose $A = \begin{bmatrix} \mathbf{a}_1^t \\ \mathbf{a}_2^t \\ \vdots \\ \mathbf{a}_m^t \end{bmatrix}$ is an $m \times n$ matrix and $\mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_m \end{bmatrix}$.

Then

- $\mathbf{c}^t A = [c_1, \dots, c_m] A = c_1 \mathbf{a}_1^t + \dots + c_m \mathbf{a}_m^t$.
- Thus, $\mathbf{c}^t A$ is a linear combination of rows of A .
- The rows of A are linearly dependent iff $\mathbf{c}^t A = \mathbf{0}^t$ (zero row) for some nonzero $\mathbf{c} \in \mathbb{R}^m$.
- Note: $\mathbf{c}^t A = \mathbf{0}$ iff $A^t \mathbf{c} = \mathbf{0}$, i.e., iff $[\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m] \mathbf{c} = \mathbf{0}$.
- Therefore, the rows of A are linearly dependent iff $\mathbf{a}_1, \dots, \mathbf{a}_m$ are linearly dependent, i.e., the columns of A^t are linearly dependent.

Result

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\} \subseteq \mathbb{R}^n$ and consider the $n \times m$ matrix $A = [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_m]$. Then S is linearly dependent *iff* the system $A\mathbf{x} = \mathbf{0}$ has a non-trivial solution.

Result

If $m > n$ then any set of m vectors in $\mathbb{R}^n$ is linearly dependent.

Result

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\} \subseteq \mathbb{R}^n$ and consider the $m \times n$ matrix $A = \begin{bmatrix} \mathbf{v}_1^t \\ \mathbf{v}_2^t \\ \vdots \\ \mathbf{v}_m^t \end{bmatrix}$. Then S is linearly dependent *iff* $\text{rank}(A) < m$.

Result

Suppose $RREF(A)$ has a zero row. Then the rows of A are linearly dependent.

Result

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\} \subseteq \mathbb{R}^n$ and $A = [\mathbf{v}_1 \ \mathbf{v}_2 \cdots \mathbf{v}_m]$. Then the following are equivalent.

- 1 S is linearly dependent.
- 2 Columns of A are linearly dependent.
- 3 $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution.
- 4 Rows of A^t are linearly dependent.
- 5 $\text{rank}(A^t) < m$.
- 6 $RREF(A^t)$ has a zero row.

Result

Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ be a linearly independent set in V and $\mathbf{v} \notin \text{span}(S)$. Then $S \cup \{\mathbf{v}\}$ is also linearly independent.

Basis: A subset B of a vector space V is said to be a **basis** for V if $\text{span}(B) = V$ and if B is **linearly independent**.

Example

$\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is a basis for $\mathbb{F}^n$. This basis is called the **standard basis** for $\mathbb{F}^n$.

Example

$\{1, x, x^2, \dots, x^n\}$ is a basis for $\mathbb{R}_n[x]$, known as the **standard basis** for $\mathbb{R}_n[x]$.

Example

$\{1, x, x^2, \dots\}$ is a basis for $\mathbb{R}[x]$, known as the **standard basis** for $\mathbb{R}[x]$.

Example

$\mathcal{E} = \{E_{11}, E_{12}, E_{21}, E_{22}\}$ is a basis for $M_2(\mathbb{R})$, where

$$E_{11} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, E_{12} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$$

$$E_{21} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \text{ and } E_{22} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Example

Is $B = \{1 + x, x + x^2, 1 + x^2\}$ a basis for $\mathbb{R}_2[x]$?

Let $p(x) = a_0 + a_1x + a_2x^2 \in \mathbb{R}_2[x]$. Then,

$$\begin{aligned} p(x) &= c_1(1 + x) + c_2(x + x^2) + c_3(1 + x^2) \\ &= (c_1 + c_3) + (c_1 + c_2)x + (c_2 + c_3)x^2 \end{aligned}$$

$$\iff c_1 = \frac{a_0 + a_1 - a_2}{2}, c_2 = \frac{a_1 + a_2 - a_0}{2}, c_3 = \frac{a_0 + a_2 - a_1}{2}$$

Ans. YES

Result

For a subspace U , a subset $B = \{\mathbf{v}_1, \dots, \mathbf{v}_r\} \subseteq U$ is a basis of U *iff* every element of U is a unique linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_r$.

Example

Find a basis for the subspace $S = \{\mathbf{x} \in \mathbb{C}^4 : A\mathbf{x} = \mathbf{0}\}$, where

$$A = \begin{bmatrix} 1 & -1 & -1 & 2 \\ 2 & -2 & -1 & 3 \\ -1 & 1 & -1 & 0 \end{bmatrix}, \quad \text{RREF}(A) = \begin{bmatrix} 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Note: $S_h = \{c_1[1, 1, 0, 0]^t + c_2[-1, 0, 1, 1]^t : c_1, c_2 \in \mathbb{C}\}$

Let $a, b, c \in \mathbb{R}$. Then

$$a(1+x) + b(x+x^2) + c(1+x^2) = 0$$

$$\Rightarrow \begin{bmatrix} 1+x & x+x^2 & 1+x^2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 1 & x & x^2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \mathbf{0}, \text{ as } \{1, x, x^2\} \text{ is LI}$$

$$\Rightarrow \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \mathbf{0}, \text{ as the rank of } \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \text{ is } 3$$

Thus, $\{1+x, x+x^2, 1+x^2\}$ is LI.

- Note the correspondence

$$1+x \longleftrightarrow \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad x+x^2 \longleftrightarrow \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad 1+x^2 \longleftrightarrow \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

- $\{1+x, x+x^2, 1+x^2\}$ is LI iff $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$ is LI.

Result

Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_m\} \subseteq V$ and $T \subseteq \text{span}(S)$ such that $n = |T| > m$. Then T is *linearly dependent*.

Proof.

Let $T = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$. For $1 \leq j \leq n$, let

$$\mathbf{u}_j = a_{1j}\mathbf{v}_1 + \dots + a_{mj}\mathbf{v}_m = [\mathbf{v}_1 \ \dots \ \mathbf{v}_m] \begin{bmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{bmatrix} = [\mathbf{v}_1 \ \dots \ \mathbf{v}_m] \mathbf{a}_j.$$

Then

$$c_1\mathbf{u}_1 + \dots + c_n\mathbf{u}_n = [\mathbf{u}_1 \ \dots \ \mathbf{u}_n] \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = [\mathbf{v}_1 \ \dots \ \mathbf{v}_m] [\mathbf{a}_1 \ \dots \ \mathbf{a}_n] \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

Now $[\mathbf{a}_1 \ \dots \ \mathbf{a}_n] \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = \mathbf{0}$ has a nontrivial solution for c_j (why?).

Therefore, $c_1\mathbf{u}_1 + \dots + c_n\mathbf{u}_n = \mathbf{0}$ has a nontrivial solution. So, T is LD.

Coordinates: Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be an **ordered** basis for a vector space $V(\mathbb{F})$ and let $\mathbf{v} \in V$. Let $\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n$. Then the scalars $c_1, c_2, \dots, c_n$ are called the **coordinates of $\mathbf{v}$** with respect to B , and the column vector

$$[\mathbf{v}]_B = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{F}^n$$

is called the **coordinate vector** of $\mathbf{v}$ with respect to B .

★ Coordinate of a vector is always associated with an **ordered** basis.

Example

The coordinate vector $[p(x)]_B$ of $p(x) = 1 - 3x + 4x^2$ with respect to basis $B = \{1, x, x^2\}$ of $\mathbb{R}_2[x]$ is $[p(x)]_B = \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix}$.

Result

Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be an ordered basis for a vector space V , let $\mathbf{u}, \mathbf{v} \in V$ and let $c \in \mathbb{F}$. Then

$$[\mathbf{u} + \mathbf{v}]_B = [\mathbf{u}]_B + [\mathbf{v}]_B \quad \text{and} \quad [c\mathbf{u}]_B = c[\mathbf{u}]_B.$$

Result

Let B be an ordered basis with n vectors for a vector space V , $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\} \subseteq V$, and $T = \{[\mathbf{u}_1]_B, [\mathbf{u}_2]_B, \dots, [\mathbf{u}_k]_B\}$. Then,

- S is a spanning set for V iff T is a spanning set for $\mathbb{F}^n$.
- S is LI in V iff T is LI in $\mathbb{F}^n$.

Result

Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis for a vector space V .

- 1 Any set of *more than n vectors* in V must be *linearly dependent*.
- 2 Any set of *fewer than n vectors* in V *cannot span* V .

Result

Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_m\} \subseteq V$. Then S contains a basis of $\text{span}(S)$.

- The vector space V is called **finite-dimensional** if V has a finite spanning set.
- Else, V is called **infinite-dimensional**.

Result (The Basis Theorem)

Every finite-dimensional vector space V **has a basis**. Further, every basis for V has equal number of vectors.

- The **dimension** of V , denoted $\dim V$, is the **number of vectors in a basis** for V .
- We write $\dim V = \infty$ if V does not have a finite basis.
- The space V is **finite-dimensional** iff $\dim V$ is a **finite** number.
- The space V is **infinite-dimensional** iff there is an **infinite** linearly independent set in V .
- The dimension of the zero space $\{\mathbf{0}\}$ is defined to be **0**.

Example

$\dim(\mathbb{R}^n) = n$, $\dim \mathbb{C}(\mathbb{C}) = 1$, $\dim \mathbb{C}(\mathbb{R}) = 2$, $\dim M_2(\mathbb{R}) = 4$ and $\dim \mathbb{R}_n[x] = n + 1$.

Result

Let V be a vector space with $\dim V = n$.

- 1 Any *linearly independent* set in V contains *at most n* vectors.
- 2 Any *spanning set* for V contains *at least n* vectors.
- 3 Any linearly independent set of *exactly n* vectors in V is *a basis* for V .
- 4 Any spanning set for V of *exactly n* vectors is *a basis* for V .
- 5 Any linearly independent set in V *can be extended to* a basis for V .
- 6 Any spanning set for V *can be reduced to* a basis for V .

Result

Let W be a subspace of a *finite-dimensional* vector space V .
Then

- 1 $\dim W \leq \dim V$;
- 2 $\dim W = \dim V$ if and only if $W = V$.

Change of Basis: Let $B = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ and $C = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be ordered bases for a vector space V . The $n \times n$ matrix whose columns are the coordinate vectors $[\mathbf{u}_1]_C, [\mathbf{u}_2]_C, \dots, [\mathbf{u}_n]_C$ is denoted by $P_{C \leftarrow B}$, and is called the **change of basis matrix** from B to C . That is,

$$P_{C \leftarrow B} = [[\mathbf{u}_1]_C, [\mathbf{u}_2]_C, \dots, [\mathbf{u}_n]_C].$$

Result

Let $B = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ and $C = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be ordered bases for a vector space V and let $P_{C \leftarrow B}$ be the change of basis matrix from B to C . Then

- 1 $P_{C \leftarrow B}[\mathbf{x}]_B = [\mathbf{x}]_C$ for all $\mathbf{x} \in V$;
- 2 $P_{C \leftarrow B}$ is the *unique matrix* P with the property that $P[\mathbf{x}]_B = [\mathbf{x}]_C$ for all $\mathbf{x} \in V$;
- 3 $P_{C \leftarrow B}P_{B \leftarrow C} = I_n = P_{B \leftarrow C}P_{C \leftarrow B}$.

Example

Find the change of basis matrix $P_{C \leftarrow B}$ and $P_{B \leftarrow C}$ for the ordered bases $B = \{1, x, x^2\}$ and $C = \{1 + x, x + x^2, 1 + x^2\}$ of $\mathbb{R}_2[x]$. Then find the coordinate vector of $p(x) = 1 + 2x - x^2$ with respect to the basis C .